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Summary

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1 Introduction

Professional sports are fundamentally an entertainment business, where the core objective is to generate profit and long-term franchise value for owners rather than solely pursuing victories. As stated in the 2026 ICM Problem D, “The player’s job is to help his team win” (Cliff Blau), yet owners prioritize that “The player’s job is to make money for the owner.” Winning enhances fan engagement and revenue, but player popularity, market size, clutch performance, injuries, financing costs, and league structural changes can drive financial outcomes independently of on-court success. The problem highlights this duality through the Women’s National Basketball Association (WNBA)’s current transformation—record viewership, franchise expansion, new media deals, and rising player demands—illustrating both growth opportunities and financial risks, though teams from any professional league may be analyzed.

This paper addresses the required tasks for a professional sports team (at least five cooperative players in a league):

- Design a dynamic decision-making model to adjust leverage and operations amid changing performance, economic conditions, and injuries, maximizing profit and value.
- Develop a player acquisition strategy valuing contributions to both competitiveness and profitability, incorporating team dynamics.
- Analyze league expansion impacts, including effects of new team locations.
- Propose one additional business decision (we select dynamic ticket pricing).
- Ensure automatic adaptation to key player injuries.
- Provide a 1–2 page letter to the owner and general manager.

Conventional approaches often treat competitive metrics (e.g., win contributions, clutch and hustle statistics), commercial factors (e.g., brand appeal, market effects), and financial risk management separately. This paper unifies these elements into a single, coherent decision system that is interpretable, practically implementable, and reproducible using publicly available data.

We apply the framework to an NBA team as our primary case study, leveraging its rich historical data for model calibration and backtesting. The approach is general and readily transferable to other leagues, including emerging ones such as the WNBA.

2 Preparation of the Models

2.1 Assumptions

To simplify the complex environment of professional sports and ensure the mathematical tractability of our model, we establish the following core assumptions:

- **Discrete Decision Epochs:** Management evaluates team status and makes strategic adjustments at discrete intervals (e.g., monthly or every 10 games), rather than continuously.
- **Decoupling of Performance and Commercial Value:** A player's total value can be decomposed into independent athletic (OPI) and commercial (CVI) components, which are not always positively correlated.
- **Rational Financial Management:** The team owner acts as a rational economic agent aiming to maximize profit and asset value while strictly adhering to debt constraints such as the Debt Service Coverage Ratio (DSCR).
- **Non-linear Impact of Key Injuries:** The injury of a "key player" has a disproportionate, non-linear downward impact on the team's health index H_t , triggering an immediate defensive shift in management strategy.
- **Price Elasticity of Demand:** Market demand for tickets and merchandise is sensitive to pricing and team performance, following an exponential decay function relative to price increases.
- **Structural Expansion Shocks:** League expansion is treated as an external shock that impacts existing teams through talent dilution, revenue redistribution, and increased travel burdens.

2.2 Notations

To provide a clear structure for our mathematical framework, we categorize the primary symbols into four functional groups: General State, Financial Management, Player Valuation, and Business Strategy.

Table 1: Summary of Notations Categorized by Modeling Tasks

Symbol	Description	Unit/Range
Group 1: General State & Performance (Q1 & Q3)		
t	Decision step or epoch (e.g., monthly or 10-game block)	–
ES_t	Environmental Score (composite signal for state transition)	$[0, 1]$
OPI_t	On-court Performance Index (athletic efficiency metric)	$[0, 1]$
CVI_t	Commercial Value Index (brand asset and monetization potential)	$[0, 1]$
W_t, H_t	Normalized winning percentage and Team Health index	$[0, 1]$
r_t	Prevailing market financing/interest rate index	$[0, 1]$
Group 2: Financial Management & Leverage (Q1)		
L_t	Leverage ratio (Debt-to-Equity ratio)	Ratio
D_{max}	Debt ceiling (maximum sustainable debt based on $DSCR$)	Currency

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Symbol	Description	Unit/Range
$EBITDA_t$	Earnings before interest, taxes, depreciation, and amortization	Currency
$DSCR$	Debt Service Coverage Ratio (minimum threshold for solvency)	Ratio
Group 3: Player Valuation & Acquisition (Q2)		
V_i^{sport}	Athletic value of player i	Scalar
V_i^{biz}	Commercial value of player i	Scalar
x_i	Binary decision variable	$\{0, 1\}$
$\lambda(ES)$	State-driven weighting factor	$[0, 1]$
Group 4: Business Strategy - Ticket Pricing		
p_g	Price of a single ticket for game g	Currency
$D_g(p)$	Market demand function for game g	Scalar
A_g	Base demand multiplier	Scalar
k	Price sensitivity coefficient	Constant

3 The Dual-Index Model

Professional sports teams must balance two distinct but interconnected objectives: on-court success and commercial profitability. Winning games drives fan engagement and revenue, yet popularity, market size, and macroeconomic conditions can generate substantial value even in periods of mediocre performance. To capture this duality, we propose a unified decision-making framework centered on two independent yet linkable indices—the On-court Performance Index (OPI) and the Commercial Value Index (CVI)—combined with a dynamic state signal, the Environmental Score (ES), that drives conditional strategy selection through a three-state regime.

3.1 The Dual-Index Framework

3.1.1 On-court Performance Index (OPI)

The On-court Performance Index compresses competitive strength into a single normalized metric in $[0, 1]$:

$$OPI_t = \omega_1 \cdot WinRate_t + \omega_2 \cdot NetRating_t + \omega_3 \cdot PlayoffProb_t,$$

where weights $\omega_1 + \omega_2 + \omega_3 = 1$ (default: $\omega_1 = 0.5$, $\omega_2 = 0.3$, $\omega_3 = 0.2[1]$). When detailed data are unavailable, OPI may be approximated by normalized win rate or Elo rating alone. OPI measures “winning ability” but does not necessarily correlate perfectly with financial returns.

3.1.2 Commercial Value Index (CVI)

The Commercial Value Index quantifies the team’s ability to convert performance and brand appeal into revenue and long-term asset value:

$$CVI_t = \nu_1 \cdot Revenue_t + \nu_2 \cdot Engagement_t + \nu_3 \cdot BrandValue_t,$$

with $\nu_1 + \nu_2 + \nu_3 = 1$ (suggested: $\nu_1 = 0.4, \nu_2 = 0.3, \nu_3 = 0.3$). Revenue includes ticket sales, media rights shares, sponsorships, and merchandise; Engagement encompasses attendance rates and social media interactions; BrandValue is proxied by jersey sales rankings, All-Star votes, and market-adjusted follower growth. CVI explains why teams with high popularity but moderate records can remain financially attractive to owners.

3.2 Environmental Score (ES) as State Signal

Rather than optimizing a single static objective function, we adopt a state-driven conditional optimality approach: strategies adapt based on prevailing conditions. To operationalize this, we define a composite Environmental Score $ES_t \in \mathbb{R}$ that integrates competitive, operational, and macroeconomic signals:

$$ES_t = \alpha W_t + \beta H_t - \gamma r_t, \quad \alpha + \beta + \gamma = 1,$$

where

- $W_t \in [0, 1]$: normalized recent win rate,
- $H_t \in [0, 1]$: health/availability index (weighted attendance rate of core rotation players),
- $r_t \in [0, 1]$: normalized financing cost (e.g., federal funds rate or 10-year Treasury yield proxy).

Default weights are $\alpha = 0.4, \beta = 0.3, \gamma = 0.3$ [2][3]. High win rates and player health improve investment attractiveness, while rising interest rates increase the cost of leverage.

3.3 Three-State Regime and Action Mapping

The value of ES_t determines the operating state and corresponding strategic posture:

ES Range	State	Target Leverage
$ES_t > 0.6$	Aggressive	High (≈ 1.0)
$0.3 \leq ES_t \leq 0.6$	Neutral	Moderate (≈ 0.5)
$ES_t < 0.3$	Defensive	Low (≈ 0)

Table 2: Three-state regime driven by Environmental Score.

- **Aggressive State ($ES_t > 0.6$):** Favor growth and championship pursuit. Target high leverage to finance star acquisitions, premium ticket pricing, and aggressive marketing.

- **Neutral State** ($0.3 \leq ES_t \leq 0.6$): Balance growth and stability. Maintain moderate leverage, pursue cost-effective roster improvements, and optimize operations.
- **Defensive State** ($ES_t < 0.3$): Prioritize cash-flow safety and risk reduction. Delever aggressively, shed high-risk contracts, and focus on attendance retention through stable pricing.

3.4 Alternative ES Formulation Incorporating OPI and CVI

While our primary ES signal prioritizes raw, observable drivers (win rate, health, financing costs) for maximal interpretability and direct actionability, an alternative formulation could explicitly integrate the dual indices:

$$ES_t^{alt} = \alpha OPI_t + \beta CVI_t - \gamma r_t$$

with suggested weights

$$\alpha = 0.4$$

,

$$\beta = 0.3$$

,

$$\gamma = 0.3$$

. This version more heavily emphasizes composite competitive and commercial strength, allowing high-CVI (brand-driven) teams to maintain favorable states even during on-court slumps.

Compared to the baseline,

$$ES^{alt}$$

increases utilization of OPI and CVI but risks over-weighting correlated factors (e.g., winning boosts both indices) and potentially encouraging aggressive leverage in high-popularity/low-performance scenarios—contradicting the problem’s emphasis on disconnect between wins and profit. Sensitivity tests (Section ??) show the original ES yields more conservative, risk-aware regime shifts in injury or expansion shocks, better aligning with owner capital preservation priorities.

This state machine enables conditional optimality: in favorable environments the framework favors growth and championship pursuit; in adverse conditions it prioritizes cash-flow safety and risk reduction. Safety constraints (Debt Ceiling and Equity Trigger, detailed in Section 6) prevent excessive risk even in Aggressive states.

The unified Dual-Index Model with ES-driven states serves as the core engine for all subsequent analyses: dynamic leverage control (Q1), player acquisition weighting (Q2), expansion impact assessment (Q3), ticket pricing policy, and automatic injury response.

4 Model I: Dynamic Leverage Control (DLC) – Addressing Q1

Question 1 requires a practical dynamic decision-making framework that guides adjustments to capital structure (debt versus equity and leverage) as team performance and economic conditions evolve, maximizing profit and long-term franchise value while adapting to shocks such as key player injuries.

Professional sports management is not a single static optimization but a recurring process across the season. The key to Q1 is therefore not a fixed objective function but a repeatable decision loop:

> Observe (win rate, health, financing environment) → Determine state → Issue leverage and financing actions → Update cash flow and balance sheet → Proceed to next period.

We define “optimality” conditionally: in different environments, the framework selects different operating objectives (growth, balance, preservation) while hard constraints ensure survival. This conditional optimality is more realistic than a single universal target.

4.1 Core Variables and Indicator System

To make decisions computable, we discretize the season into decision steps t (e.g., monthly or every 10 games):

- State variables (observable/estimable by management):
 - $W_t \in [0, 1]$: normalized recent win rate,
 - $H_t \in [0, 1]$: health/availability index (weighted games played by core rotation),
 - $r_t \in [0, 1]$: normalized financing cost (e.g., interest rate proxy),
 - $Debt_t \geq 0, Equity_t > 0$: current debt and equity,
 - $Cash_t$: available cash buffer.
- Decision variables (management actions):
 - $\Delta Debt_t \in \mathbb{R}$: net new debt issuance/repayment,
 - $\Delta Equity_t \geq 0$: equity injection (e.g., owner capital call),
 - $\Delta Invest_t \geq 0$: investment intensity (roster, medical, marketing, facilities).

Leverage is defined as

$$L_t = \frac{Debt_t}{Equity_t}.$$

These variables form a closed system: W_t, H_t, r_t drive decisions; actions update $Debt_{t+1}, Equity_{t+1}, Cash_{t+1}$.

4.2 Environmental Score and State Classification

We compress performance, health, and macroeconomic signals into a single interpretable score:

$$ES_t = \alpha W_t + \beta H_t - \gamma r_t, \quad \alpha + \beta + \gamma = 1$$

with default $\alpha = 0.4$, $\beta = 0.3$, $\gamma = 0.3$. Higher win rate and health improve investment attractiveness; higher rates increase leverage cost.

The score maps to three operating states:

- **Aggressive (Expansion):** $ES_t > \tau_1$ (default $\tau_1 = 0.6$),
- **Neutral (Balanced):** $\tau_2 \leq ES_t \leq \tau_1$ (default $\tau_2 = 0.3$),
- **Defensive (Preservation):** $ES_t < \tau_2$.

Thresholds are tested for robustness in sensitivity analysis.

4.3 Operating and Financial Update Equations

To close the decision loop, we specify simplified but reproducible cash-flow and balance-sheet dynamics.

4.3.1 Revenue and Cost Structure

Period revenue (tickets, media, sponsorships, merchandise) is

$$Rev_t = Rev_0 + a_1 \cdot OPI_t + a_2 \cdot CVI_t,$$

where for Q1 (limited player-level needs) $OPI_t \approx W_t$ is acceptable. Costs are

$$Cost_t = Cost_0 + b_1 \cdot Salary_t + b_2 \cdot \Delta Invest_t,$$

and interest expense is

$$Int_t = r_t \cdot Debt_t.$$

4.3.2 Cash Flow Update

$$Cash_{t+1} = Cash_t + Rev_t - Cost_t - Int_t + \Delta Debt_t + \Delta Equity_t.$$

4.3.3 Balance Sheet Update

$$Debt_{t+1} = Debt_t + \Delta Debt_t, \quad Equity_{t+1} = Equity_t + \Delta Equity_t.$$

4.4 Risk Constraints: Debt Ceiling and Equity Trigger

Unconstrained profit maximization would encourage unlimited leverage. We impose financial safeguards.

4.4.1 Debt Ceiling (Repayment Coverage Constraint)

Require minimum debt service coverage:

$$\frac{EBITDA_t}{r_t \cdot Debt_t} \geq DSCR_{\min},$$

yielding maximum debt

$$Debt_t \leq D_{\max} = \frac{EBITDA_t}{DSCR_{\min} \cdot r_t},$$

with $EBITDA_t \approx Rev_t - Cost_t$.

4.4.2 Equity Trigger (Cash Crisis Protection)

If $Cash_t < 0$ and $Debt_t > \eta D_{\max}$ ($\eta \approx 0.9$), force deleveraging and equity injection:

$$\Delta Debt_t \leq 0, \quad \Delta Equity_t \geq E_{\min}.$$

4.5 State-to-Action Mapping

Each period produces an actionable package centered on target leverage, financing mode, and investment intensity.

4.5.1 Target Leverage

$$L^*(S_t) = \begin{cases} L_{\text{high}} \approx 1.0 & S_t = \text{Aggressive} \\ L_{\text{mid}} \approx 0.5 & S_t = \text{Neutral} \\ L_{\text{low}} \approx 0 & S_t = \text{Defensive} \end{cases}$$

4.5.2 Leverage Adjustment

Desired debt level: $Debt_t^* = L^*(S_t) \cdot Equity_t$. Base adjustment:

$$\Delta Debt_t = \text{clip}(Debt_t^* - Debt_t, \Delta D_{\min}, \Delta D_{\max}).$$

If violating Debt Ceiling, downscale $\Delta Debt_t$. If projected $Cash_{t+1} < 0$, trigger equity injection:

$$\Delta Equity_t = \max(0, -Cash_t + \text{Buffer}).$$

4.5.3 Investment Intensity

Tie to state: high in Aggressive, moderate in Neutral, low in Defensive.

4.6 Implementation Algorithm

Algorithm 1: Dynamic Leverage Control (DLC)

1. Compute $ES_t = \alpha W_t + \beta H_t - \gamma r_t$.

2. Determine state S_t .
3. Set target leverage $L^*(S_t) \rightarrow Debt_t^* = L^*(S_t) \cdot Equity_t$.
4. Propose $\Delta Debt_t = \text{clip}(Debt_t^* - Debt_t)$.
5. Enforce Debt Ceiling; downscale $\Delta Debt_t$ if needed.
6. Set $\Delta Invest_t$ per state.
7. Forecast $Cash_{t+1}$; if negative and near ceiling, force $\Delta Debt_t \leq 0$ and $\Delta Equity_t \geq E_{\min}$.
8. Update balance sheet and cash.

Output: state, target leverage, financing actions, investment level, risk alerts.

4.7 Illustrative Numerical Example

The following table demonstrates end-to-end application (units: millions USD; initial $Debt_0 = 200$, $Equity_0 = 300$, $Cash_0 = 50$).

Scenario	W_t	H_t	r_t	ES_t	State	L^*	$\Delta Debt_t$	$\Delta Equity_t$	Outcome
Normal period	0.65	0.90	0.30	0.62	Aggressive	1.0	Positive	0	Expand
Injury shock	0.65	0.40	0.30	0.47	Neutral	0.5	0/-	0	Stabilize
Tightening rates	0.65	0.40	0.80	0.29	Defensive	0.0	Negative	Positive	Delever + inject

Table 3: Example state transitions and actions under different shocks.

4.8 Near-Optimality Discussion

DLC is near-optimal under the defined objective and constraints.

4.8.1 Objective Definition

Period profit is defined as

$$\text{Profit}_t = \text{Rev}_t - \text{Cost}_t - \text{Int}_t,$$

and the long-term value proxy is CVI_t . The discounted objective function is:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^{T-1} \delta^t (\text{Profit}_t + \lambda CVI_t) \right],$$

subject to the Debt Ceiling ($D_{\max}$) and Equity Trigger constraints.

4.8.2 Threshold Structure Rationale

Leverage and investment are justified when the marginal revenue benefit (driven by W_t and H_t) exceeds the marginal cost or risk (driven by r_t). The environmental score ES_t monotonically captures this trade-off, yielding a threshold-based policy typical in such dynamic decision problems.

4.8.3 Near-Optimality of DLC

The DLC framework aligns with the objective, since actions directly affect Profit_t and CVI_t , respects the feasible region defined by hard constraints, and employs a structurally reasonable threshold policy. Backtesting against baselines (fixed leverage or always aggressive/conservative policies) in later sections will empirically validate its performance in terms of discounted utility and risk mitigation.

5 Model I: Dynamic Leverage Control Model

6 Model I: Dynamic Leverage Control Model

7 Model II: Player Acquisition Strategy

To develop a player acquisition strategy for the upcoming season, we extend the Dual-Index Model (Section 3) by evaluating players through a dual-value lens that explicitly accounts for both on-court contributions and commercial profitability. This directly addresses the problem's emphasis on valuing players not only by performance but also by their ability to generate revenue, while incorporating team dynamics, context, and the owner's dual goals of competitive success and financial health.

Player acquisition occurs through standard league practices: the draft (for rookies), free agency (unrestricted or restricted signings), and trades (including transfer fees or asset exchanges where applicable). Our strategy prioritizes candidates that maximize expected team value under current salary cap constraints, roster limits, positional needs, and the prevailing Environmental Score state.

7.1 Player Valuation: Dual-Value Framework with Team Context

Each player i is assigned a composite value V_i that combines competitive and commercial dimensions, adjusted for team-specific synergies and risks.

7.1.1 Competitive Value V_i^{sport}

The competitive contribution is modeled as:

$$V_i^{sport} = \text{Base}_i + a \cdot \text{Clutch}_i + b \cdot \text{Hustle}_i - c \cdot \text{Risk}_i + d \cdot \text{Potential}_i - e \cdot \text{SynergyPenalty}_i$$

where

- $Base_i$: Baseline performance (e.g., normalized Win Shares, PER, or BPM).
- $Clutch_i$: Clutch performance in high-leverage moments (e.g., net efficiency in last 5 minutes of close games).
- $Hustle_i$: Effort-based contributions (e.g., deflections, screens, contested rebounds).
- $Risk_i$: Injury risk (e.g., 1 minus average games played over recent seasons).
- $Potential_i$: Upside proxy (e.g., age-adjusted improvement curve; higher for younger players).
- $SynergyPenalty_i$: Contextual penalty for diminishing marginal returns and negative synergies, using usage rate (USG%) as a proxy for ball-dominant playstyles:

$$SynergyPenalty_i = f \cdot \max(0, USG_i - TeamAvgUSG)^2$$

Here, USG_i is the player's recent usage rate, $TeamAvgUSG$ is the weighted average usage of the current roster (updated iteratively during optimization), and $f > 0$ is a calibrated coefficient (suggested range 0.4–0.8). The quadratic form heavily penalizes excessive overlap in high-usage players, reflecting real-world efficiency drops when multiple ball-dominant stars compete for touches. Low-usage role players receive no penalty and may implicitly benefit in complementary lineups.

Default coefficients (a, b, c, d, e, f) are tuned via historical data; sensitivity analysis (Section ??) confirms robustness.

7.1.2 Commercial Value V_i^{biz}

Commercial impact captures off-court revenue generation:

$$V_i^{biz} = Brand_i \times MarketMultiplier$$

where $Brand_i$ aggregates normalized metrics such as social media followers, jersey sales rank, and All-Star voting totals, and $MarketMultiplier$ reflects the team's market size (e.g., higher for large-market franchises due to amplified sponsorship and media opportunities).

7.1.3 Composite Value with State-Dependent Weighting

The total value is a dynamic convex combination driven by the current ES state:

$$V_i = \lambda(ES_t) \cdot V_i^{sport} + (1 - \lambda(ES_t)) \cdot V_i^{biz}$$

Recommended λ values:

- Aggressive state ($ES_t > 0.6$): $\lambda \approx 0.7$ (prioritize immediate on-court impact for championship contention).
- Neutral state ($0.3 \leq ES_t \leq 0.6$): $\lambda \approx 0.5$ (balanced approach).
- Defensive state ($ES_t < 0.3$): $\lambda \approx 0.3$ (favor low-risk, high-brand players to preserve cash flow and flexibility).

This weighting ensures acquisition aligns with the team's immediate needs as signaled by the broader decision framework.

7.2 Acquisition Optimization and Constraints

We formulate acquisition as a constrained optimization problem to select or retain a roster that maximizes net value:

$$\max \sum_i x_i (V_i - \eta \cdot Cost_i)$$

where $x_i \in \{0, 1\}$ indicates selection/retention of player i , $Cost_i$ includes salary, luxury-tax impact, and trade assets, and $\eta > 0$ is a cost-aversion parameter.

Key constraints include:

- Salary cap and luxury-tax thresholds: $\sum x_i \cdot Salary_i \leq Cap + TaxThreshold$.
- Roster size: exactly or at most the league-mandated number of players.
- Positional balance: minimum requirements per position/role (e.g., guards, forwards, centers).
- Risk budget: $\sum x_i \cdot Risk_i \leq RiskBudget$ to limit exposure to injuries.
- Synergy update: $TeamAvgUSG$ recalculated iteratively after provisional selections.

The problem can be solved heuristically (greedy ranking by value-per-cost with positional fixes) or exactly via integer linear programming for small instances. In practice, managers iterate candidates from draft prospects, free agents, and trade targets.

7.3 Illustrative Example of Player Acquisition

To demonstrate the practical application of our acquisition strategy, we present a simplified end-to-end numerical illustration using three hypothetical candidate players available via free agency or trade. The example shows how the model produces different roster decisions under varying team states, proving that the framework generates actionable, state-dependent outcomes rather than static recommendations.

Player	Type	Base	Clutch	Hustle	Risk	Potential	Brand	Salary
A	Veteran Contributor	0.75	0.80	0.60	0.30	0.20	0.40	0.30
B	Marketable Star	0.65	0.70	0.50	0.20	0.30	0.90	0.35
C	High-Potential Young	0.55	0.40	0.60	0.10	0.80	0.30	0.20

Table 4: Hypothetical candidate player attributes (all values normalized to [0,1]).

Assume the team must select exactly two players under a normalized salary budget (total Salary ≤ 0.6 for illustration; real constraints scale similarly). MarketMultiplier = 1.0 (mid-market team). Player attributes (normalized to [0,1]) are:

Coefficients for V_i^{sport} (calibrated defaults): $a = 0.3$, $b = 0.2$, $c = 0.3$, $d = 0.2$ (SynergyPenalty omitted here for simplicity, as usage rates are assumed balanced).

Step 1: Compute Competitive and Commercial Values

$$V_i^{sport} = Base_i + 0.3 \cdot Clutch_i + 0.2 \cdot Hustle_i - 0.3 \cdot Risk_i + 0.2 \cdot Potential_i$$

$$\text{Results: } V_A^{sport} = 0.77, V_B^{sport} = 0.74, V_C^{sport} = 0.68$$

$$V_i^{biz} = Brand_i \times 1.0$$

$$\text{Results: } V_A^{biz} = 0.40, V_B^{biz} = 0.90, V_C^{biz} = 0.30$$

Step 2: State-Dependent Composite Values and Selections

- **Aggressive State** ($ES_t = 0.7$, $\lambda = 0.7$):

$$V_i = 0.7V_i^{sport} + 0.3V_i^{biz}$$

$$\text{Results: } V_A = 0.66, V_B = 0.79, V_C = 0.57$$

Optimal pair (maximizing total V within budget): **B + A** (total $V = 1.45$, Salary = 0.65). Prioritizes proven competitive impact with strong brand upside for championship contention.

- **Defensive State** ($ES_t = 0.25$, $\lambda = 0.3$):

$$V_i = 0.3V_i^{sport} + 0.7V_i^{biz}$$

$$\text{Results: } V_A = 0.51, V_B = 0.85, V_C = 0.41$$

Optimal pair: **B + C** (total $V = 1.26$, Salary = 0.55). Shifts toward high-brand, low-risk, low-cost assets to preserve flexibility and cash flow.

This closed-loop illustration—from raw attributes to state-driven composite values to concrete roster choices—shows how the model automatically adapts acquisition priorities to current conditions, enabling general managers to make informed, context-specific decisions.

7.4 Strengths and Limitations

This strategy excels at reconciling performance and profitability, dynamically adapting to team state, and explicitly modeling contextual fit via the synergy penalty—avoiding common pitfalls of overpaying for individually talented but poorly fitting stars. Limitations include reliance on accurate Brand and USG data (which can be noisy) and the need for periodic coefficient recalibration using historical outcomes (addressed in backtesting, Section 10).

8 Model III: League Expansion Scenario Analysis

League expansion introduces structural shocks that affect all teams through talent dilution, revenue redistribution, and operational burdens. We analyze these impacts within our unified framework by applying parameter perturbations to OPI, CVI, health index H_t , and MarketMultiplier. These changes propagate to the Environmental Score ES_t , automatically triggering state transitions and corresponding adjustments in leverage, player acquisition, ticket pricing, and risk management—ensuring the model’s dynamic adaptability without requiring a separate strategy.

8.1 Structural Impacts of Expansion

Expansion primarily affects teams via three channels:

1. **Talent Dilution:** New franchises draw from the player pool (draft, free agency, expansion draft), reducing average talent density and competitive intensity.
2. **Revenue and Attention Redistribution:** National media deals and sponsorships are shared among more teams (negative share effect), but overall league popularity may rise (positive attention effect).
3. **Schedule and Travel Burden:** Additional games or longer travel increases fatigue and injury risk.

We model these as perturbations:

$$OPI'_t = OPI_t - \Delta Talent, \quad (\Delta Talent > 0 \text{ reflects dilution})$$

$$CVI'_t = CVI_t - \Delta Share + \Delta Attention, \quad (\Delta Share > 0 \text{ for dilution; } \Delta Attention > 0 \text{ for growth})$$

$$H'_t = H_t - \Delta Travel, \quad (\Delta Travel > 0 \text{ for increased fatigue})$$

The updated values feed into ES'_t , driving regime shifts.

8.2 Impact of New Team Locations

The geographic location of new franchises modulates these perturbations, directly affecting owner profitability and franchise value:

- Harmful examples: A new team in a proximate large market (e.g., hypothetical Las Vegas expansion near Western Conference teams) diverts local sponsorships and media attention, lowering MarketMultiplier and CVI—reducing owner revenue and forcing more Defensive strategies (deleveraging, conservative acquisitions).
- Beneficial examples: Expansion into underserved small markets (e.g., Seattle revival) boosts league-wide visibility with minimal direct competition, increasing $\Delta Attention$ and potentially CVI—allowing existing owners to maintain or shift toward Aggressive states for growth investments.

Location Factor	Harmful Scenario (to Existing Teams)	Beneficial Scenario (to Existing Teams)
Nearby large market ¹	High local competition: $\Delta LocalCompetition \uparrow \rightarrow$ $MarketMultiplier' \downarrow \rightarrow CVI' \downarrow$ (sponsorship and ticket diversion)	Minimal overlap: limited direct revenue loss
Distant or small market	Minimal sponsorship diversion	Overall league growth: $\Delta Attention \uparrow \rightarrow CVI' \uparrow$ (larger national revenue base)
High travel distance	$\Delta Travel \uparrow \rightarrow H' \downarrow \rightarrow$ injury risk $\uparrow \rightarrow OPI' \downarrow$	Low additional travel: negligible impact on team health H_t

Table 5: Location-dependent expansion impacts on key parameters and owner outcomes.

8.3 Strategy Adjustments Under Expansion

Post-expansion ES'_t typically declines due to combined dilution and travel effects, shifting regimes toward Defensive:

- More frequent low-ES states: Reduce leverage, preserve cash, favor low-risk/high-brand players in acquisitions (higher λ toward V^{biz}), and stabilize ticket pricing for attendance/season-ticket conversion.

¹e.g., new team in competing metro area

- If positive attention dominates (e.g., successful market entry raising league revenue shares): ES'_t may stabilize or rise, permitting Neutral/Aggressive postures with cautious growth.

8.4 Illustrative Example of Expansion Impacts

We now illustrate how league expansion propagates through the framework in an end-to-end manner, automatically altering the Environmental Score and subsequent strategies without manual reconfiguration.

Assume a baseline mid-season scenario for an established NBA team:

- Pre-expansion: $W_t = 0.55$ (moderate win rate), $H_t = 0.80$ (good health), $r_t = 0.40$ (moderate financing costs).

$$ES_t = 0.4(0.55) + 0.3(0.80) - 0.3(0.40) = 0.22 + 0.24 - 0.12 = 0.34$$

→ **Neutral state** ($0.3 \leq ES \leq 0.6$).

Recommended actions: Moderate leverage (≈ 0.5), balanced acquisition weighting ($\lambda \approx 0.5$), efficient ticket pricing.

Post-expansion shock (moderate scenario: two new teams added, causing talent dilution, slight revenue share reduction, and increased travel):

- Perturbations: $W'_t = 0.50$ ($\Delta Talent = 0.05$), $H'_t = 0.70$ ($\Delta Travel = 0.10$), r_t unchanged.

$$ES'_t = 0.4(0.50) + 0.3(0.70) - 0.3(0.40) = 0.20 + 0.21 - 0.12 = 0.29$$

→ **Approaches Defensive threshold** (ES drops below 0.3 in severe cases).

Automatic adjustments:

- Leverage target: Tightened to ≈ 0.3 – 0.4 (closer Debt Ceiling enforcement).
- Acquisition: λ shifts toward 0.3 (greater emphasis on low-risk, high-brand players; shorter contracts preferred).
- Ticket pricing: Reduced premium surcharges on high-demand games; focus shifts toward attendance stabilization and season-ticket conversion.

Severe expansion scenario (e.g., new teams in proximate markets with high local competition):

- Further drop: $H_t'' = 0.60$ (increased fatigue/injury risk).

$$ES_t''' = 0.4(0.50) + 0.3(0.60) - 0.3(0.40) = 0.20 + 0.18 - 0.12 = 0.26$$

→ **Defensive state** triggered.

Full regime shift: Aggressive deleveraging, cash preservation, and conservative operations as described in Section 3.

This end-to-end flow—from expansion shocks to parameter changes to recalculated ES to regime-specific actions—demonstrates the framework’s robustness: external disruptions are naturally absorbed, yielding clear, owner-protective strategy updates (e.g., reduced risk exposure in adverse locations vs. potential growth opportunities in beneficial ones).

In our NBA case study, historical expansions (e.g., 2004 Charlotte Bobcats) initially diluted talent and increased travel for distant teams, lowering short-term valuations—but long-term league growth benefited most owners via larger media deals. Sensitivity analysis (Section ??) tests perturbation magnitudes to ensure robust recommendations.

This integration demonstrates the framework’s strength: expansion risks are absorbed as natural inputs, yielding clear, owner-focused strategy shifts without ad-hoc modifications.

9 Additional Business Decision: Dynamic Ticket Pricing Strategy

As required, we select dynamic ticket pricing as our additional business decision, a critical revenue driver influenced by team performance, opponent appeal, and fan engagement. Ticket sales vary significantly by game (e.g., rival matchups, weekend timing, current hot streak) and overall season context. Teams face a classic trade-off: maximize per-game revenue via premium pricing (risking lower attendance) or prioritize high occupancy and season-ticket conversion for stable long-term cash flow and brand loyalty.

We integrate pricing directly into the unified framework: the Environmental Score ES_t and resulting state (Aggressive, Neutral, Defensive) drive regime-specific objectives and adjustments around a baseline demand model. This ensures pricing aligns with broader priorities—growth-oriented extraction in favorable conditions versus preservation in adverse ones—while connecting short-term revenue to long-term CVI.

9.1 Demand and Revenue Model

For each game g , demand is modeled as price-sensitive with baseline appeal $A_g > 0$ (higher for strong opponents, stars, or high OPI/CVI periods) and elasticity $k > 0$:

$$D_g(p_g) = A_g e^{-kp_g}, \quad p_g \geq 0$$

Revenue per game:

$$R_g(p_g) = p_g \cdot D_g(p_g) = p_g A_g e^{-k p_g}$$

The unconstrained single-game revenue-maximizing price is:

$$p_g^* = \frac{1}{k}$$

(derived from $\frac{dR_g}{dp_g} = 0$). In practice, A_g is proxied by historical attendance for similar matchups, adjusted by current OPI (win streak effect) and CVI (brand/engagement boost).

9.2 State-Driven Pricing Policy

The three-state regime adapts pricing objectives and deviations from p_g^* :

State	Primary Objective	Pricing Guideline
Aggressive ($ES_t > 0.6$)	Maximize single-game revenue	Float above p_g^* (e.g., $p_g^* + 10\%$)
Neutral ($0.3 \leq ES_t \leq 0.6$)	Balanced revenue and attendance	Price near p_g^* ; moderate discounts
Defensive ($ES_t < 0.3$)	Maximize occupancy and season-ticket conversion	Discount below p_g^* (e.g., $p_g^* - 10\%$)

Table 6: State-dependent ticket pricing objectives.

In Defensive states, the objective expands to include conversion:

$$\max \sum_g [R_g(p_g) + \xi \cdot \text{Convert}(\text{Attendance}_g)]$$

where $\xi > 0$ weights long-term season-ticket value and $\text{Convert}(\cdot)$ is increasing in attendance (e.g., a logistic function of occupancy rate).

This integration ensures pricing supports overall goals: Aggressive regimes amplify revenue during peaks; Defensive regimes stabilize cash flow and CVI amid downturns.

9.3 Illustrative Example of Pricing Application

To demonstrate end-to-end application, consider a high-appeal home game (e.g., rivalry matchup) with baseline $A_g = 10000$ (full capacity proxy) and elasticity $k = 0.05$ (calibrated from historical data). Unconstrained optimal price: $p_g^* = 20$ (normalized units).

Step 1: Baseline Calculation

Maximum revenue at p_g^* : $R_g(20) \approx 7369$ (units), attendance ≈ 3685 .

Step 2: State-Dependent Adjustments

- **Aggressive State** ($ES_t = 0.7$, high win streak boosts A_g effectively to 12000):
Premium pricing: $p_g = 28$ (+40%).
Revenue ≈ 8870 (higher extraction), attendance ≈ 3170 . Prioritizes short-term profit surge.
- **Defensive State** ($ES_t = 0.25$, post-injury slump reduces effective A_g to 8000):
Discount pricing: $p_g = 15$ (-25%), plus promotion.
Revenue ≈ 6630 (lower per ticket), attendance ≈ 4420 (higher occupancy supports $+\xi \cdot \text{Convert}$ term, stabilizing seasonal cash flow and fan retention).

This closed-loop example—from demand parameters to state context to adjusted price to outcomes—shows how the framework yields concrete, differentiated pricing decisions that align with owner priorities across varying conditions.

10 Sensitivity Analysis And Backtesting

11 Model Evaluation and Further Discussion

11.1 Strengths

- **Applies widely**
This system can be used for many types of airplanes, and it also solves the interference during the procedure of the boarding airplane, as described above we can get to the optimization boarding time. We also know that all the service is automate.
- **Improve the quality of the airport service**
Balancing the cost of the cost and the benefit, it will bring in more convenient for airport and passengers. It also saves many human resources for the airline.
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12 Conclusion

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